

Optimal Execution under Market Impact

Real Data Execution and Volume-Aware Extensions

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Abstract

We extend optimal execution analysis from simulated environments to real intraday market data. Using empirical price and volume series, we evaluate execution strategies under realistic conditions and introduce a volume-aware extension of the Almgren–Chriss model. The results highlight the importance of liquidity in execution performance and demonstrate that hybrid strategies outperform both purely model-based and purely volume-based approaches.

1 Introduction

Previous analysis evaluated execution strategies under stochastic price simulation. In this study, we transition to a real market setting, where:

- Prices S_k are directly observed
- Volume V_k is empirically measured
- Execution occurs under realistic market microstructure

The objective is to evaluate whether theoretical optimal execution models remain effective under real data.

2 Real Market Framework

We consider a discrete execution process:

$$x_k = Q - \sum_{j=1}^k n_j \tag{1}$$

where:

- Q = total shares to execute
- n_k = shares executed at time k

- x_k = remaining inventory

Execution prices incorporate market impact:

$$P_k = S_k + \gamma \sum_{j < k} n_j + \eta \frac{n_k}{\tau} \quad (2)$$

where:

- γ = permanent impact
- η = temporary impact
- τ = time step

3 Empirical Volume Profile

In contrast to the simulated setting, execution weights are derived from real market volume:

$$w_k = \frac{V_k}{\sum_{i=1}^N V_i} \quad (3)$$

This enables VWAP to align execution with observed liquidity.

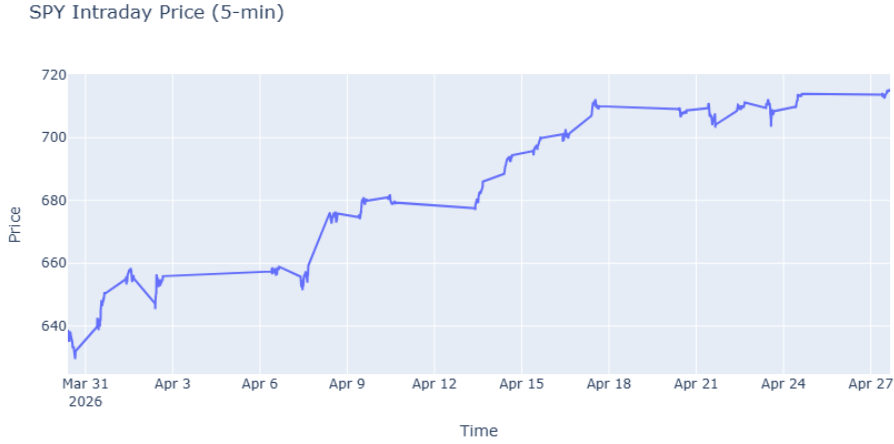


Figure 1: SPY intraday price (5-minute intervals)

4 Execution under Real Data

Execution strategies were applied using real prices and empirical volume.

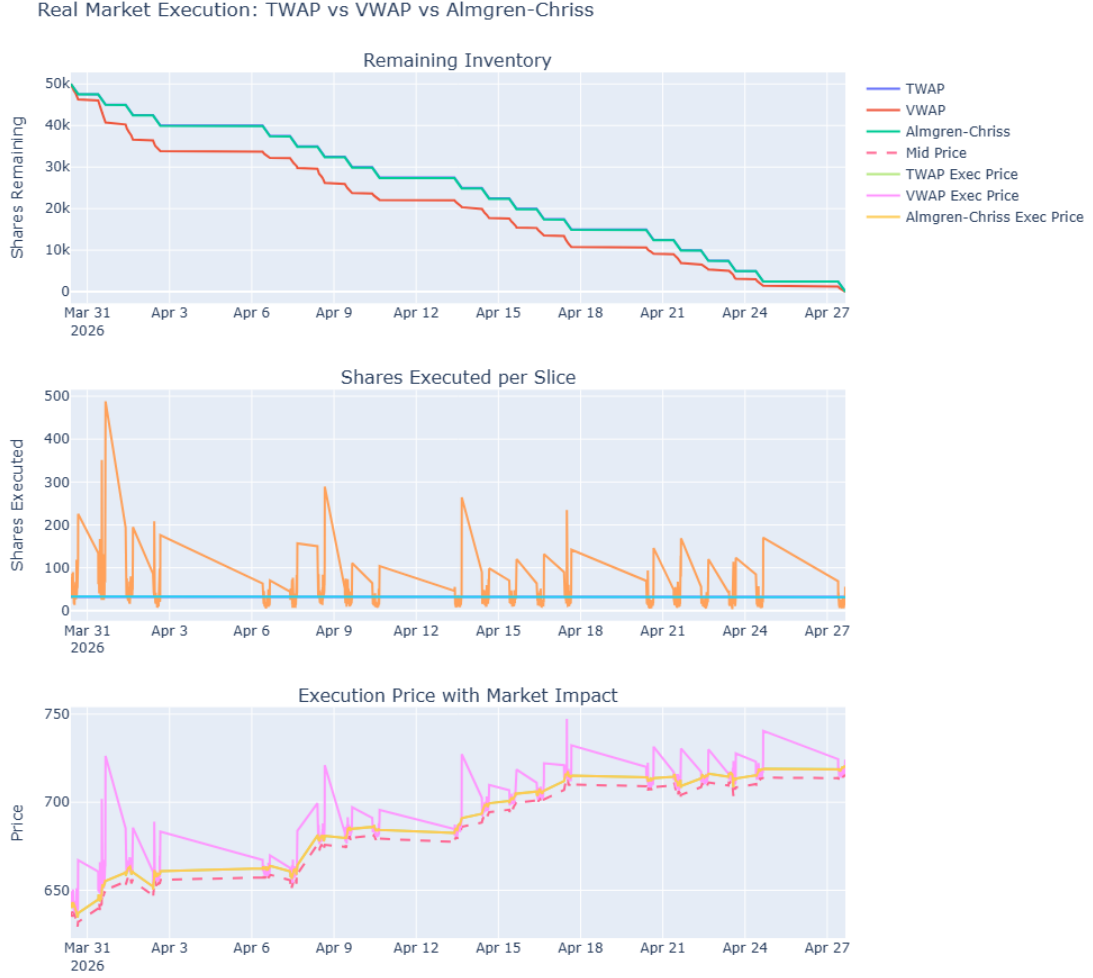


Figure 2: Execution comparison under real data

Observation:

- VWAP achieves lower execution cost due to liquidity alignment
- TWAP provides a stable baseline
- Almgren-Chriss behaves similarly to TWAP under low λ

5 Lambda Calibration

We calibrate λ using:

$$\lambda^* = \arg \min_{\lambda} \text{Risk}(\lambda) \quad (4)$$

subject to:

$$\text{Cost}_\lambda \leq \text{Cost}_{VWAP}(1 + \delta) \quad (5)$$

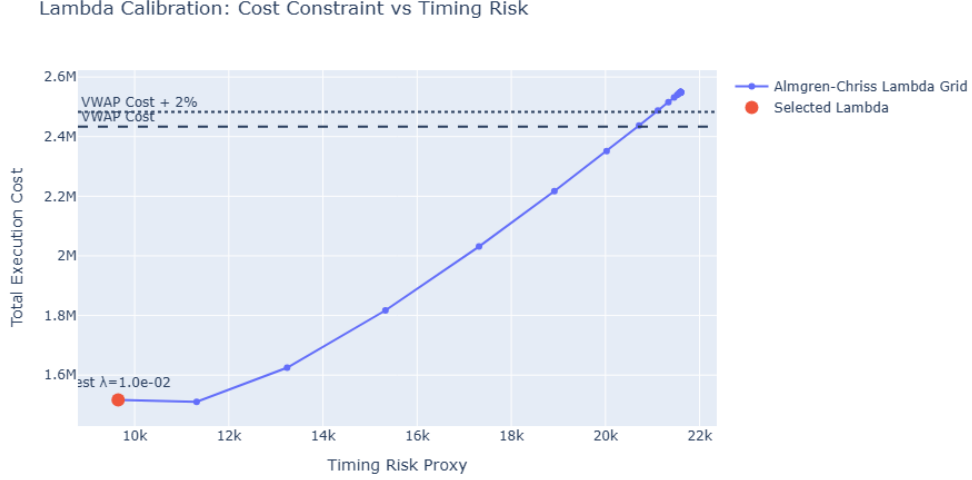


Figure 3: Lambda calibration: cost vs risk

Result:

The standard Almgren–Chriss model does not satisfy the cost constraint relative to VWAP, indicating that:

- VWAP dominates in cost under real market conditions
- Almgren–Chriss is highly sensitive to parameter calibration

6 Volume-Aware Almgren–Chriss

To incorporate real market liquidity, we define a hybrid execution strategy:

$$n_k^{VA} = \alpha n_k^{AC} + (1 - \alpha) n_k^{VWAP} \quad (6)$$

where:

- n_k^{VA} = volume-aware execution at time k
- n_k^{AC} = Almgren–Chriss execution schedule
- n_k^{VWAP} = volume-weighted execution schedule
- $\alpha \in [0, 1]$ = blending parameter

Interpretation:

- $\alpha = 1 \rightarrow$ pure Almgren–Chriss
- $\alpha = 0 \rightarrow$ pure VWAP
- $0 < \alpha < 1 \rightarrow$ hybrid execution strategy

This formulation allows the execution strategy to balance:

- Risk control (from Almgren–Chriss)
- Liquidity awareness (from VWAP)



Figure 4: Volume-aware execution: cost vs risk

Key Result:

$$\alpha^* \approx 0.2 - 0.3 \quad (7)$$

This produces lower execution cost than both VWAP and standard Almgren–Chriss.

7 Key Insights

- Liquidity timing is the dominant driver of execution cost
- VWAP outperforms model-based strategies when volume is informative
- The standard Almgren–Chriss model underperforms when ignoring market microstructure
- Hybrid strategies outperform both pure approaches

Main Insight:

The best execution strategy is not purely theoretical nor purely data-driven, but a combination of both.

8 Conclusion

The results demonstrate that real market conditions fundamentally alter execution performance.

- Optimal execution models must incorporate liquidity information
- Market microstructure plays a central role in execution outcomes
- Hybrid approaches provide a practical improvement over classical methods

This motivates future work on adaptive execution strategies where decisions depend dynamically on market state variables.

9 References

- Almgren, R., & Chriss, N. (2001). Optimal execution of portfolio transactions.
- Cartea, Á., Jaimungal, S., & Penalva, J. (2015). Algorithmic and High-Frequency Trading.